// CROSS AI SOFTWARES INC. · EXECUTIVE ARCHITECTURE

NextGen Architecture Blueprint

Residential Property Intelligence Operating System — Stratex Core · Stratex Passport · Stratex Habitat. Executive review package for Phase 1 authorization.

STATUS · DRAFT · AWAITING EXECUTIVE APPROVAL

VERSION 1.0 · July 18, 2026 · CONFIDENTIAL · CLASSIFICATION: BLACK-OPS

STRATEX™ · A CROSS AI SOFTWARES INC. PRODUCT

PREPARED BY TC · AI PROD

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STRATEX CORE — NEXTGEN ARCHITECTURE BLUEPRINT

Version: 1.0 · **Status**: DRAFT · Awaiting Executive Approval

Author: TC · Stratex AI Product Manager · Under directive of Anthony Cross

Directive Reference: STRATEX CORE DIRECTIVE 001 — Legacy Freeze & NextGen Rebuild

Design principle: *Optimize for correctness of architecture, not speed of completion.*

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0. EXECUTIVE FRAMING

Stratex Core NextGen is **not** a drone-inspection app.

It is the **Residential Property Intelligence Operating System** — an enterprise-grade

mission-control platform that orchestrates hardware, AI, human review, and permanent

data custody across three fused pillars:

PILLAR	PURPOSE
Stratex Core	Contractor operating system — mission, capture, intelligence, delivery
Stratex Passport	Permanent, hash-chained property intelligence record — the source of truth
Stratex Habitat	Homeowner-facing intelligence platform

The drone is **one sensor among many**. The AI workforce performs the intelligence.

The Property Passport is the permanent source of truth. Habitat is the homeowner face.

Mandatory architectural workflow — no module bypasses this chain:

```
Mission Control → Mission Planning → Mission Validation → Flight →
Mission Assurance → Capture Validation → Digital Twin Generation →
AI Workforce Processing → Property Intelligence → AWE™ Intelligence →
Human QA → Report Generation → Property Passport Update →
Habitat Synchronization → Customer Delivery
```

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1. APPLICATION STATE MODEL

Three formally separated states co-exist in the codebase:

STATE	LOCATION	DEPLOY TARGET	MODIFIABLE?	PURPOSE
PRODUCTION	`stratexdrone.com`	Frozen at last approved deploy	No — read-only	Customer-visible; unchanged during rebuild
NEXTGEN	`stratex-quant.preview.emergentagent.com`	Active dev	**Yes** — all new work	Rebuild target
LEGACY	`/legacy/*` namespace inside preview	Not deployable to prod	Read-only after freeze	Rollback, comparison, knowledge preservation

Migration rules — irreversible until parity

Legacy is **never deleted** until ALL of the following pass:

1. Replacement complete
2. QA passed
3. APIs migrated
4. Data migration verified
5. Passport sync verified
6. Habitat sync verified

7. Reports validated
8. Navigation validated
9. AI workflows validated
10. Regression suite passed

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2. MODULE HIERARCHY

The NextGen platform is organized into **14 first-class modules** grouped under 4 layers.

Layer I — Operations (mission execution)

1. **Mission Control** — command deck; live global state; alerts; fleet status
2. **Mission Planning** — waypoint design, mission templates, geofencing, altitude profiles
3. **Mission Validation** *(formerly Air Traffic Control)* — 20-item preflight hard gate; blocks launch on any failure
4. **Flight Operations** — live telemetry, breadcrumbs, in-air observability
5. **Mission Assurance** — in-flight QA agent; monitors capture density, GSD, overlap; requests additional shots before landing
6. **Capture Validation** — post-flight ingestion; radiometric integrity checks; RGB/thermal registration verification

Layer II — Intelligence (data → insight)

1. **Digital Twin Studio** — 3-model workspace (Reality Capture · Engineering CAD/BIM · Property Intelligence)
2. **AI Workforce** — orchestration console for the ~20 specialized agents (see §5)
3. **Property Intelligence** — canonical property record view; systems (roof, walls, windows, doors, HVAC, ventilation)
4. **AWE™ Intelligence** *(new pillar)* — **A**ir · **W**ater · **E**nergy — cross-cutting environmental performance analytics
5. **Human QA** — review + approval console; every AI finding requires human sign-off before it enters the Passport

Layer III — Delivery

1. **Report Studio** — configurable enterprise deliverables; every value carries provenance chip
2. **Property Passport** — immutable hash-chained ledger; native module today; extractable to standalone service in Phase 5
3. **Habitat Sync** — outbound sync of approved data to the homeowner platform

Layer IV — Platform

- **Identity & Access** — auth, roles, org tenancy, audit
- **Integrations** — DJI, weather, FAA airspace, materials/labor DBs, payments
- **System Console** — health, keys, feature flags, sync queues

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3. NAVIGATION HIERARCHY (canonical)

Single persistent operator shell. No hidden orphans.

```
┌─ MISSION CONTROL ─────────────────────────────────
│  · Overview
│  · Active Missions
│  · Alerts & Notifications
│  · Fleet Status
│
├─ PROPERTY PORTFOLIO ─────────────────────────────
│  · Properties (all)
│  · New Property
│  · Property Detail
│     ├─ Overview
│     ├─ Digital Twin
│     ├─ AWE™ Intelligence
│     ├─ Findings
│     ├─ Reports
│     ├─ Passport Ledger
│     └─ Timeline
│
├─ MISSIONS ────────────────────────────────────────
│  · Mission Planner
│  · Mission Validation (ATC)
│  · Flight Operations (live)
│  · Mission Assurance (in-flight QA)
│  · Capture Validation (post-flight)
│  · Mission History
│
├─ DIGITAL TWIN STUDIO ────────────────────────────
│  · Reality Capture Layer
│  · Engineering CAD/BIM Layer
│  · Property Intelligence Layer
│  · Layer Sync & Compare
│  · Historical Diff
│
├─ AI WORKFORCE ────────────────────────────────────
│  · Agent Registry
│  · Agent Runs (queue + history)
│  · Confidence & Evidence Console
│  · Prompt Version Manager
│  · Cost Ledger
│
├─ AWE™ INTELLIGENCE ──────────────────────────────
│  · Air (ventilation, air quality, leakage)
│  · Water (moisture, roof intrusion, wall retention)
│  · Energy (thermal, envelope, performance score)
│  · AWE Composite Score
│
├─ ESTIMATING ──────────────────────────────────────
│  · Materials Catalog
│  · Labor Catalog
│  · Regional Pricing
│  · National Pricing
│  · Contractor Custom Pricing
│  · Estimate Builder
│  · Estimate Versions
│
├─ REPORTING ───────────────────────────────────────
│  · Report Templates
│  · Report Studio (WYSIWYG)
│  · Delivered Reports
│  · Report Versions
│
├─ HUMAN QA ────────────────────────────────────────
│  · Approval Queue
│  · Rejection Log
│  · QA Audit Trail
│
├─ INTEGRATIONS ────────────────────────────────────
│  · Drone Fleet (DJI)
│  · Weather (Open-Meteo)
│  · Airspace (FAA · LAANC — AWAITING INTEGRATION)
│  · Materials/Labor DBs — AWAITING INTEGRATION
│  · Passport (native)
│  · Habitat (AWAITING INTEGRATION)
│
├─ SYSTEM ──────────────────────────────────────────
│  · Users & Roles
│  · Organizations & Tenants
│  · Security (keys, MFA, audit)
```

```
│  · Sync Queues & Health
│  · Feature Flags
│  · Settings
│
└─ LEGACY (hidden, read-only) ─────────────────────
   · /legacy/* — all pre-NextGen screens for reference
```

Every route below the top-level module MUST be reachable from the persistent left nav

OR from a canonical entity detail view. **No orphaned URLs.**

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4. DATABASE DOMAINS

The current single-tenant, dict-based Mongo layer is replaced by **10 typed domains**, each

with Pydantic document models, `to_mongo() / from_mongo()` helpers, and enforced

`created_at / updated_at / created_by / org_id` audit fields.

Domain 1 · Identity & Access

COLLECTION	PURPOSE
`orgs`	Multi-tenant root
`users`	Human accounts
`roles`	RBAC definitions
`sessions`	JWT session ledger
`api_keys`	Machine tokens
`audit_events`	Every mutation with actor + before/after

Domain 2 · Property Registry

COLLECTION	PURPOSE
`properties`	Canonical property record (parcel, address, geo)
`addresses`	Normalized street/city/state/zip
`parcels`	County-parcel linkage
`owners`	Legal + occupant ownership records
`contractors`	Contractor profiles (replaces scattered `_contractor` dicts)
`property_org_bindings`	Which orgs may access which properties

Domain 3 · Mission Operations

COLLECTION	PURPOSE
`missions`	Formal mission record
`mission_plans`	Waypoints, altitude, camera plans
`mission_validations`	ATC 20-item results
`mission_telemetry`	In-flight breadcrumbs
`capture_sessions`	Post-flight ingest records

Domain 4 · Capture Assets (object storage)

COLLECTION	PURPOSE
`assets`	Metadata for RGB, thermal, video, twin exports; body in S3-compatible store
`asset_derivatives`	Downsamples, PNG previews, cropped tiles
`asset_provenance`	Chain-of-custody per asset

Domain 5 · Intelligence Findings

COLLECTION	PURPOSE
`findings`	Every AI/human finding — one row per anomaly / measurement / classification
`finding_evidence`	Links to source assets + regions
`finding_confidence`	Model + confidence + version + provenance
`agent_runs`	Every AI invocation with input hash, output, cost, model version

Domain 6 · AWE™ Metrics

COLLECTION	PURPOSE
`awe_readings`	Per-inspection Air/Water/Energy readings
`awe_scores`	Composite AWE score per property/inspection
`awe_history`	Longitudinal AWE trend per property

Domain 7 · Estimating

COLLECTION	PURPOSE
`materials_catalog`	SKU, unit, unit price by region
`labor_catalog`	Trade, rate, region
`estimates`	Estimate documents linked to inspections
`estimate_versions`	Immutable revisions

Domain 8 · Reports & Deliverables

COLLECTION	PURPOSE
`report_templates`	Versioned templates
`reports`	Rendered report records
`report_versions`	Every render with provenance snapshot
`report_signatures`	Human approvals

Domain 9 · Passport & Habitat Sync

COLLECTION	PURPOSE
`passport_ledger`	Hash-chained immutable ledger (existing, migrated + typed)
`passport_sync_log`	Outbound sync audit trail
`habitat_sync_log`	Habitat-bound events
`sync_retry_queue`	Failed sync retries with backoff

Domain 10 · Compliance & Observability

COLLECTION	PURPOSE
`provenance_chain`	Every value's derivation lineage
`event_log`	Immutable event bus record
`error_events`	Exceptions with correlation IDs
`cost_ledger`	AI + storage + compute cost per mission/property

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5. AI WORKFORCE ARCHITECTURE

Every agent conforms to a strict contract:

```
class Agent(Protocol):
    id: str                # e.g. "roof-intel-v2"
    version: str           # semver
    inputs: JsonSchema     # required + typed
    outputs: JsonSchema    # always includes: value, confidence, evidence_ids, provenance
    model: str             # e.g. anthropic/claude-sonnet-4-6
    prompt_version: str    # hash of the prompt template
    cost_estimate_usd: float
    human_gate_required: bool
```

Agent registry

#	AGENT	RESPONSIBILITY	CONFIDENCE SOURCE	HUMAN GATE
1	**Mission Commander**	Compose + validate mission plan	Rule-based + Claude	Yes
2	**Mission QA**	ATC 20-item gate	Deterministic checks	No (all-or-nothing)
3	**Flight Operations**	Live telemetry watch, safe-return decisions	Rule-based	Escalates to human
4	**Mission Assurance**	In-flight capture density QA	GSD/overlap math	No (auto-retake)
5	**Capture Validation**	Radiometric integrity, RGB/thermal registration	Statistical + Claude	Yes
6	**Photogrammetry**	Point-cloud + mesh generation	AWAITING INTEGRATION	Yes
7	**Thermal Analysis**	Radiometric anomaly detection	Model + statistical	Yes
8	**CAD/BIM Generation**	Structural layers, framing extraction	AWAITING INTEGRATION	Yes
9	**Digital Twin**	Fuse 3 layers into single twin	Deterministic composition	Yes
10	**Roof Intelligence**	Roof plane, facet, damage classification	CV + Claude	Yes
11	**Window Intelligence**	Window type, dimensions, glass, leakage	CV + Claude	Yes
12	**Door Intelligence**	Door type, material, seal condition	CV + Claude	Yes
13	**Water (AWE)**	Moisture retention, roof water risk	Thermal + Claude	Yes
14	**Air (AWE)**	Ventilation performance, leakage paths	Thermal + Claude	Yes
15	**Energy (AWE)**	Envelope performance, heat-loss score	Thermal + Claude	Yes
16	**Damage Detection**	Anomaly clustering across all systems	CV + Claude	Yes
17	**Measurement Intelligence**	Extract precise measurements from twin	Deterministic geometry	Yes
18	**Materials Intelligence**	Compose BOM with confidence	Catalog + Claude	Yes
19	**Labor Intelligence**	Compose labor plan	Catalog + Claude	Yes
20	**Estimating Intelligence**	Final estimate with waste, tax, O&P	Deterministic math	Yes
21	**Maintenance Priority**	Rank issues by urgency + cost avoidance	Composite scoring	Yes
22	**Reporting Intelligence**	Compose enterprise report	Template + Claude narrator	Yes
23	**Passport Sync**	Emit ledger delta to Passport service	Deterministic	No
24	**Habitat Sync**	Emit homeowner packet to Habitat	Deterministic	No

Absolute rules (non-negotiable)

1. **No agent may fabricate a measurement.** If input is insufficient, output must be `null` + reason.
2. **Every output carries `confidence_pct`, `evidence_ids[]`, `provenance{}`.**
3. **Every human-gated finding is BLOCKED from entering the Passport until QA approves.**
4. **Every agent run is written to `agent_runs`** with input hash, output, cost, model version, prompt version — permanent audit.
5. **Cost caps per property + org** — hard-stop when exceeded.

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6. API BOUNDARIES

Five formal API planes, each with its own auth, rate limits, and versioning.

PLANE	PREFIX	AUDIENCE	AUTH
Public Read API	`/api/pub/v1/`	Anonymous (via passport hash)	Signed URL / hash
Product API	`/api/v1/`	Authenticated users	JWT + org context
Agent API	`/api/agents/v1/`	Backend agent runners only	Service token
Sync API	`/api/sync/v1/`	Passport + Habitat services	mTLS + service token
Webhook API	`/api/hooks/v1/`	External inbound (DJI, FAA, weather)	HMAC signature

All existing routes at *`/api/passport/`*, *`/api/claim-snapshot/`*, *`/api/mission-control/*`*

are preserved under legacy compatibility shims that forward to the new plane 1:1 during

the parity window.

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7. EVENT FLOW & STATE MANAGEMENT

Event-driven backbone. Every domain mutation emits an immutable event to `event_log`.

Downstream consumers subscribe by event type.

```
mission.planned          → validates mission
mission.validated        → unlocks flight
flight.completed         → triggers capture_validation
capture.validated        → schedules AI workforce runs
finding.produced         → notifies human_qa
finding.approved         → passport.ledger.append
passport.updated         → habitat.sync.request
```

Frontend state: **React Query** for server state + **Zustand** for local UI state.

Rejected — Redux (too much ceremony), Recoil (deprecating), unmanaged local state (current problem).

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8. SYNCHRONIZATION FLOW

```
STRATEX CORE  ─(approved finding)→  PASSPORT LEDGER  ─(sync webhook)→  HABITAT
                                          │
                                          └─(sync webhook)→  CARRIER (optional)
```

- **Idempotency**: every sync carries a UUID; duplicate reception is a no-op
- **Ordering**: strict per-property via sequence numbers
- **Retry**: exponential backoff to `sync_retry_queue`
- **Audit**: every attempt logged to `passport_sync_log` / `habitat_sync_log`
- **Cryptographic receipt**: SHA-256 over the payload, written into the ledger

(already implemented for the Claim Snapshot co-sign — reuse the pattern)

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9. SECURITY MODEL

Identity

- JWT (short-lived) + refresh tokens
- MFA required for CEO + admin roles
- SSO (SAML / OIDC) — AWAITING INTEGRATION (Phase 5)

Authorization

- **RBAC**: CEO · GM · Contractor · Inspector · Pilot · Homeowner · Adjuster · Service
- **ABAC overlay**: property-level access via `property_org_bindings`
- Every API call enforces: `authorize(user, action, resource)`

Data protection

- Passport ledger: hash-chained (already implemented)
- Cryptographic receipts on all critical writes (already implemented for co-sign)
- Field-level encryption on PII (address, owner name, contact) using envelope encryption
- Object storage: signed URLs with expiry; per-org bucket isolation

Provenance chain

Every value shown to a user must resolve to a `provenance{}` record answering:

1. **Source** — sensor, agent, human, calculation
2. **Confidence** — 0-100 or "verified"
3. **Version** — model version, prompt version
4. **Timestamp** — captured, computed, approved
5. **Signer** — if human-approved, who + when

Observability

- Sentry (frontend + backend) — AWAITING INTEGRATION
- Structured logs (JSON, trace-id correlated)
- Distributed tracing via OpenTelemetry — AWAITING INTEGRATION
- Health probes: `/healthz`, `/readyz`, `/livez`

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10. DEPLOYMENT MODEL

Three environments, promoted linearly:

```
NextGen preview  →  Staging  →  Production (stratexdrone.com)
```

- Every deploy requires: passing `iter-XX` regression suite + human sign-off
- **Blue/green** at the Emergent ingress layer
- **Rollback**: existing Emergent checkpoint feature (no code needed)
- **Feature flags**: LaunchDarkly or open-source alternative — AWAITING INTEGRATION
- **CI/CD**: pushed to Emergent-managed pipeline; supplemented by repo-side lint/test workflow — AWAITING INTEGRATION

Object storage (S3-compatible) required for asset domain — **AWAITING INTEGRATION**.

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11. VISUAL DESIGN PRINCIPLES

Not negotiable. Every screen must pass this rubric:

Aesthetic pillars

1. **Mission-control system**, not a form-flow app
2. **Dark graphite base** — never white primary
3. **Glass panels** — 12-24px backdrop blur, subtle stroke
4. **Teal illumination** as anchor color (`#00E5FF`)
5. **Orange highlights** as accent (`#FF7B00`)
6. **Gold ceremonial** for approvals (`#D4B86A`)
7. **Magenta for urgency** (`#FF2D78`)
8. **Green for validated** (`#00FF9C`)
9. **Typography**: Sora display · Space Grotesk body · JetBrains Mono for data
10. **Generous negative space** — 2-3× typical SaaS density
11. **Cinematic motion** — reveal choreography, no boring transitions
12. **Provenance chip on every meaningful value** (mandatory shared component)

Anti-patterns (forbidden)

- Purple/violet gradients on white
- Generic card grids with equal padding
- Emoji as functional icon (allowed only in written narrative)
- Placeholder Lorem Ipsum
- Any dashboard tile that shows a mock number without a `MOCK` chip

Provenance chip specification

Every displayed value shall carry one of the following badges:

- 🟢 **MEASURED** — from a physical sensor
- 🔵 **CALCULATED** — from measured inputs
- 🟡 **ESTIMATED** — from a model with confidence <95%
- ✅ **VERIFIED** — human QA approved
- 🤖 **AI-ASSISTED** — model output pending human QA
- ⏳ **AWAITING INTEGRATION** — capability shipping in a future phase
- 🎭 **DEMO DATA** — visible in prototype builds only

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12. EXECUTIVE ARCHITECTURE REVIEW

Per your directive: *"Do not assume the architecture is perfect. Critique it."*

Strengths

1. **Passport hash chain is genuinely defensible** — cryptographic tamper-evidence is rare in this industry
2. **Cosign SHA-256 receipt pattern is reusable** across every human sign-off in the platform
3. **Storm→calendar bridge + Open-Meteo integration** proves the platform can react to real-world events
4. **PDF pipeline is production-grade** — Playwright + cache-first design already survives Emergent K8s
5. **Emergent LLM Key + `emergentintegrations`** eliminates vendor lock-in on AI

Weaknesses

1. **Monolithic `server.py`** (2,684 lines) — will not survive scale; must be broken into per-domain routers
2. **No canonical property model** — the single `property_passports` document does too much
3. **Zero product-route auth** — the biggest deployment risk today

4. **No object storage** — PDFs and images live on ephemeral disk
5. **AI orchestration is ad-hoc** — no agent registry, no cost ledger, no prompt versioning
6. **No frontend TypeScript** — 50 pages of unchecked JS across a growing platform
7. **No test coverage on the UI** — regressions are only caught by the testing subagent

Technical debt (top-priority)

- Break `server.py` (P0)
- Add `require_auth` to every product route (P0)
- Introduce Pydantic doc models per collection (P0)
- Add object storage (P1)
- Formal event log (P1)
- Prompt version manager (P1)
- Frontend TypeScript migration (P2)
- OpenTelemetry tracing (P2)

Risks

1. **DJI SDK is closed** — the Matrice/Manifold/Dock integration will require an NDA and a licensed developer account. Timeline unknown.
2. **Photogrammetry / CAD-BIM** — no open-source pipeline meets enterprise quality; likely commercial (Bentley ContextCapture, Pix4D, Reality Capture). Cost pressure.
3. **Real thermal ingest** — radiometric R-JPG parsing needs FLIR/DJI SDK. Not free.
4. **Multi-tenant refactor** — every existing query must be rewritten. High regression risk.
5. **Ledger migration** — moving the existing Bingham passport into normalized rows must preserve every hash — mistakes here are unrecoverable.

Missing integrations (formally declared)

- FAA LAANC airspace
- DJI Cloud API + FlightHub 2
- Manifold 3 edge compute
- Photogrammetry pipeline
- CAD/BIM generator
- Materials + labor DB (regional pricing)
- Object storage (S3-compatible)
- Sentry / OpenTelemetry
- Habitat platform (its own service)
- SSO (SAML / OIDC)

Performance concerns

1. `/api/storms/active` is O(N passports × 1 external HTTP call) — must be batched + cached
2. Passport ledger unbounded growth — no compaction strategy
3. PDF cache is not shared across replicas — will race on scale-out
4. No pagination in list endpoints — will fail beyond 1k records

Security concerns

1. Public passport URL leaks all passport data to anyone with the hash — must move to signed URLs
2. Cosign token generation has no per-passport race guard
3. No CSRF on state-changing endpoints
4. WebSocket accepts any client — needs authenticated handshake

Scalability concerns

1. Single Mongo instance, no replica set — no HA
2. Single uvicorn worker — no horizontal scaling
3. Static file serving from same process as API — must move to CDN

Recommended improvements

1. **Event sourcing** for the intelligence pipeline — every finding derives from an immutable event chain
2. **CQRS split** — reads from denormalized projections, writes to normalized commands
3. **Agent-as-a-service** pattern — every agent runs as an idempotent HTTP endpoint
4. **Feature flags** for every AWAITING INTEGRATION capability so we can flip them on without deploys
5. **Contract-first API design** — publish OpenAPI specs before implementation for every new endpoint

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13. PHASE 1 DELIVERABLE MAP (traceability)

Mapping this blueprint to the seven Phase 1 acceptance criteria you set:

CRITERION	DELIVERABLE	SECTION
1. Deployable NextGen Preview	Working `/nextgen/*` route tree with shell + workspace stubs on preview URL	Implementation phase
2. Architecture Blueprint	**This document**	§1-§12
3. System Flow Validation	Workflow chain diagram + per-stage stub route	§0 workflow + §3 nav
4. Visual Design Review	Screenshots of every workspace in the shell	Implementation phase
5. AI Workforce Map	24-agent registry table	§5
6. Executive Architecture Review	Strengths · weaknesses · debt · risks · improvements	§12
7. Final deliverable	Preview + all above docs bundled + sign-off form	Post-implementation

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14. WHAT MUST NOT BE BUILT YET (Phase 1 forbidden list)

To maintain architectural correctness over speed, these are **explicitly deferred**:

- ❌ Any DJI SDK integration
- ❌ Any photogrammetry pipeline
- ❌ Any real radiometric thermal ingest
- ❌ Any CAD/BIM generator
- ❌ Real materials/labor DB
- ❌ Habitat platform itself (only the sync surface)
- ❌ SSO
- ❌ Payments live checkout
- ❌ Multi-tenant migration (deferred to Phase 2)

Every one of the above is labeled **AWAITING INTEGRATION** in the UI. No exceptions.

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15. IMPLEMENTATION SEQUENCE (once blueprint is approved)

Phase 1a — Legacy Freeze (1 session)

1. Create `/app/frontend/src/legacy/` and move every existing page there
2. Create `/app/backend/routes/legacy/` mirroring same
3. Prefix all legacy routes with `/legacy/*`
4. Delete `Andrew` page (✅ done)
5. Snapshot preserved as read-only reference

Phase 1b — NextGen Shell (1-2 sessions)

1. `/app/frontend/src/nextgen/` root

2. Persistent operator shell (top command bar, left nav, provenance-aware page frame)
3. All 14 modules as route stubs — every workspace exists, most show "AWAITING INTEGRATION"
4. Provenance chip component (shared)
5. Cross AI + Stratex logo lockup in the shell

Phase 1c — Workspace Stubs (1-2 sessions)

1. Mission Control · Property Portfolio · Missions · Digital Twin Studio · AI Workforce · AWE™ Intelligence · Estimating · Reporting · Human QA · Integrations · System
2. Every workspace has: hero, workflow-step indicator, provenance chips wired to real backend state (using existing endpoints)
3. Live workflow chain visible from any workspace

Phase 1d — Screenshot + Executive Review Package (final)

1. Screenshot every workspace on desktop + iPhone
2. Bundle blueprint + screenshots + review as a single deliverable
3. Request executive approval before Phase 2

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16. APPROVAL BLOCK

Executive approval required before Phase 1b begins.

□ Blueprint reviewed by Anthony Cross

□ Executive Architecture Review reviewed

□ Phase 1 acceptance criteria confirmed

□ Authorized to proceed to Phase 1a implementation

Signature: ______________________ Date: ______________

STRATEX CORE — NEXTGEN ARCHITECTURE REVIEW APPENDIX

Companion document to NEXTGEN_ARCHITECTURE_BLUEPRINT.md v1.0

Status: Answering Executive Review Questions 1-14

Author: TC · Stratex AI Product Manager, under directive of Anthony Cross

Purpose: Detailed responses to executive architecture review questions issued during the Blueprint hold.

Q1 — TWENTY-FOUR-AGENT ARCHITECTURE (justification)

Correction of framing

The blueprint's "24 agents" is a **logical software role count**, not a deployable-service count. In production, these logical agents collapse into **5 deployable services** during early phases:

SERVICE	LOGICAL AGENTS INSIDE	DEPLOYMENT
Mission Ops Service	Mission Commander · Mission QA · Flight Operations · Mission Assurance · Capture Validation	1 container, sync + background
Vision Intelligence Service	Photogrammetry · CAD/BIM · Digital Twin · Roof · Window · Door · Damage Detection · Measurement · Thermal Analysis	1 container, GPU-optional, async job queue
AWE Intelligence Service	Water · Air · Energy	1 container, async job queue
Estimating Service	Materials · Labor · Estimating · Maintenance Priority	1 container, sync
Delivery & Sync Service	Reporting · Passport Sync · Habitat Sync	1 container, sync + retry queue

Rationale: separately deploying 24 microservices during the prototype phase would introduce

network complexity, deployment cost, and observability burden with no operational payoff.

Per-agent detail table

#	AGENT	RESPONSIBILITY	INPUTS	OUTPUTS	MODEL / SERVICE	LOGICAL VS DEPLOYED	SYNC/ASYNC	HUMAN GATE	COST TIER
1	Mission Commander	Compose + validate mission plan	property_id, target_systems	mission_plan.json	Claude Sonnet + rule engine	Logical inside Mission Ops	Sync	Yes (dispatcher)	Low
2	Mission QA	Enforce 20-item preflight gate	mission_plan, drone_state, weather	gate_result{pass,fail,blockers[]}	Deterministic checks	Logical	Sync	No (all-or-nothing)	Free
3	Flight Operations	Live telemetry watch; safe-return decisions	telemetry stream	control_decisions[]	Rule engine	Logical	Async streaming	Escalates	Low
4	Mission Assurance	In-flight GSD/overlap QA	live imagery metadata	additional_shots[]	Statistical + Claude	Logical	Async streaming	No (auto-retake)	Low
5	Capture Validation	Radiometric integrity, RGB↔thermal registration	asset bundle	validation_report	Statistical + Claude	Logical	Async job	Yes	Low
6	Photogrammetry	Point cloud + mesh generation	RGB assets	.glb, .obj, point_cloud	**AWAITING INTEGRATION** (Pix4D / RealityCapture / OpenDroneMap)	Deployed later	Async job	Yes	High
7	Thermal Analysis	Radiometric anomaly detection	thermal R-JPG assets	thermal_findings[]	Statistical + Claude	Logical	Async job	Yes	Medium
8	CAD/BIM Generation	Structural layers, framing extraction	point cloud + mesh	.ifc, .rvt derivatives	**AWAITING INTEGRATION** (commercial)	Deployed later	Async job	Yes	High
9	Digital Twin	Fuse 3 layers into single twin	reality mesh + CAD + intelligence	twin manifest	Deterministic composition	Logical	Async	Yes	Low
10	Roof Intelligence	Plane, facet, damage classification	RGB + twin	roof_findings[]	Claude + CV heuristics	Logical	Async	Yes	Medium
11	Window Intelligence	Type, dimensions, glass, leakage	RGB + thermal + twin	window_findings[]	Claude + CV	Logical	Async	Yes	Medium
12	Door Intelligence	Type, material, seal condition	RGB + thermal + twin	door_findings[]	Claude + CV	Logical	Async	Yes	Medium
13	Water (AWE)	Moisture retention, water risk	thermal + roof geometry + weather history	water_findings[], water_score	Claude + rules	Logical inside AWE Service	Async	Yes	Medium
14	Air (AWE)	Ventilation performance	thermal + door/window findings	air_findings[], air_score	Claude + rules	Logical	Async	Yes	Medium
15	Energy (AWE)	Envelope performance, heat-loss score	thermal + envelope + weather	energy_findings[], energy_score	Claude + rules	Logical	Async	Yes	Medium
16	Damage Detection	Cross-system anomaly clustering	all findings[]	prioritized_damage[]	Claude	Logical	Async	Yes	Medium
17	Measurement Intelligence	Extract precise measurements from twin	twin + registration	measurements[] with tolerance	Deterministic geometry	Logical	Sync	Yes	Free
18	Materials Intelligence	Compose BOM	measurements + roof/window/door findings	bom[]	Catalog lookup + Claude	Logical inside Estimating	Sync	Yes	Low

#	AGENT	RESPONSIBILITY	INPUTS	OUTPUTS	MODEL / SERVICE	LOGICAL VS DEPLOYED	SYNC/ASYNC	HUMAN GATE	COST TIER
19	Labor Intelligence	Compose labor plan	bom + regional rates	labor_plan	Catalog lookup + Claude	Logical	Sync	Yes	Low
20	Estimating Intelligence	Final estimate w/ waste, tax, O&P	bom + labor + preferences	estimate.json	Deterministic math	Logical	Sync	Yes	Free
21	Maintenance Priority	Rank issues by urgency + cost avoidance	all findings + estimate	priorities[]	Composite scoring	Logical	Sync	Yes	Free
22	Reporting Intelligence	Compose enterprise report	approved intelligence bundle	report.pdf + report.json	Template + Claude narrator	Logical inside Delivery	Sync	No (post-approval)	Low
23	Passport Sync	Emit ledger delta to Passport	approved bundle	sync receipt	Deterministic	Logical	Async + retry queue	No	Free
24	Habitat Sync	Emit homeowner packet to Habitat	approved bundle	sync receipt	Deterministic	Logical	Async + retry queue	No	Free

Combination rationale

Agents 13-15 (AWE) cannot be combined because Water, Air, and Energy each require distinct

input signals (weather history vs door/window vs envelope) and produce independently-reported scores.

Similarly, Roof/Window/Door intelligences cannot be combined because approval authority

differs (a contractor may approve roof findings but require an engineer for structural doors).

Q2 — PROPERTY PASSPORT AUTHORITY

Ownership model (definitive)

CONCERN	AUTHORITY
Creates Passport	Stratex Passport service — issued via `POST /passport/v1/mint` with a nonce from Core
Owns canonical identifier	Stratex Passport service — 12-char uppercase hash, globally unique
May append records	**Stratex Passport service only.** Core submits deltas via `/passport/v1/append`; Passport service re-verifies chain integrity before writing
May correct records	Nobody. Records are immutable. Corrections are new appends with an explicit `supersedes` pointer to the prior record
Versioned via	Sequential ledger `seq` + SHA-256 chain (`prev_hash` field) — already implemented today for the native passport
Findings superseded via	New ledger entry of type `SUPERSEDE` carrying `prior_seq`, `reason`, and `superseding_finding_id`. Historical record is never destroyed
Core submission path	`POST /passport/v1/append {passport_id, delta_bundle, core_signature}` → Passport verifies + writes + returns cryptographic receipt
Habitat receipt path	`GET /passport/v1/projections/homeowner/:passport_id` — a read-only projection scoped to homeowner-appropriate fields
Conflict resolution	**First-writer-wins with per-property sequence numbers.** Concurrent appends fail with `409 SEQUENCE_CONFLICT`, forcing the loser to re-read and re-submit
Cryptographic proof	Every accepted append returns a SHA-256 receipt computed over `{payload, seq, prev_hash, signed_at}`. Core stores this receipt in its `sync_audit_log`
Deployment	**Standalone service** post-parity. Today it is embedded in `/app/backend/routes/passport.py` — this is the first module extracted in Phase 2

Access-control matrix

ACTOR	READ	APPEND	CORRECT	HABITAT RECEIVE
Contractor (Core)	✅ (their org's properties)	✅ (approved bundles only)	❌	❌
Homeowner	✅ (their property only, homeowner projection)	❌	❌	✅
Adjuster	✅ (via magic link)	✅ (cosign entries only — already exists)	❌	❌
Passport service itself	✅	✅ (all writes go through it)	✅ (via SUPERSEDE semantics)	—
Habitat service	❌ direct	❌	❌	✅ pull only

Existing Claim Snapshot cosign receipt pattern (§8 blueprint) is the canonical template for how every future append works.

Q3 — PROVENANCE INTERFACE

Visual-noise avoidance rules

Always visible (chip in-line) — applies to high-consequence numbers only:

- Envelope Score
- Moisture %
- Repair Estimate $
- AWE™ scores (Air, Water, Energy)
- Truth Score
- Confidence %

Progressive disclosure — for supporting values:

- Every measurement (dimension, count, area) shows a small colored dot only
- Tap / hover reveals a hover-card with the full provenance object
- "Open Evidence" button in the hover-card drops into an Evidence Drawer

Mobile behavior:

- Chips shrink to a single colored dot (4px) on ≤430 px viewports
- Long-press opens the provenance drawer (not hover)
- Drawer takes 90% of screen height, dismissible with swipe-down

Evidence Drawer contents:

- Source asset thumbnail (with a "view full size" affordance)
- Model + version + prompt version
- Confidence, timestamp, signer (if human-approved)
- "Reject / Escalate" affordance if the user has QA role

Provenance vs Confidence vs Truth Score (clarified)

CONCEPT	QUESTION IT ANSWERS
Provenance	*Where did this value come from?* — Source lineage
Confidence	*How sure is the model / measurement?* — Per-agent numeric or human-verified
Finding confidence	Aggregate confidence across all inputs for a single finding
Truth Score	*How trustworthy is this entire property record?* — See Q4

Q4 — TRUTH SCORE (honest specification)

Definition (revised, non-marketing)

Truth Score is a **composite property-record trust indicator**, expressed on a 0-100 scale in **10-point bands** (never in decimal form) with descriptive language:

BAND	LANGUAGE
90-100	**Field-verified · production-grade**
70-89	**AI-assisted · human-approved**
50-69	**AI-assisted · pending human review**
30-49	**Partial coverage**
0-29	**Insufficient evidence**

Component inputs

1. **Evidence coverage** (0-30 pts) — fraction of expected asset types present (RGB, thermal, dimensional)
2. **Model confidence average** (0-25 pts) — mean confidence of all findings
3. **Human verification ratio** (0-25 pts) — % of findings signed by human QA
4. **Recency** (0-10 pts) — age of freshest scan
5. **Cross-registration** (0-10 pts) — RGB↔thermal alignment score (if AWE performed)

Weighting

Deterministic weighted sum with published coefficients — never a model output.

Missing-evidence behavior

Missing evidence subtracts from Evidence Coverage. Never treated as "confident zero."

Contradictory-evidence behavior

Contradictions between two findings trigger a **CONTRADICTION** event that:

- caps Truth Score at 60 until reconciled
- surfaces both findings to the Human QA queue

Human verification effect

Each human-approved finding contributes to the Human Verification Ratio component. A property with 100% human approval on all findings can reach the 90-100 band.

Versioning

Truth Score has its own version number (e.g. `TS.v1.0`). Historical scores are frozen in the passport ledger.

Calibration

Requires empirical field-verification runs against ground truth before we publish the score externally. Until then, the score is displayed with the label **"Prototype scoring — not calibrated to industry standard yet."**

Display language rules

- **Prohibited**: "97.43%," "99.7% accurate," "AI-verified truth."
- **Required**: 10-point band + descriptive label + provenance chip

Marketing restrictions

- Truth Score may **not** be presented to insurance carriers or in contracts as a warranty of accuracy until calibration is published.
- May be presented **only** as a "record-completeness signal" during the prototype phase.

The four levels distinguished

LEVEL	SCOPE
Model confidence	Per-inference (one AI call)
Measurement confidence	Per-measurement (one dimension)
Finding confidence	Per-finding (composite of contributing model + measurement confidences)
Truth Score	Per-property record (composite of all findings + coverage + verification)

Q5 — AWE™ MISSION SEPARATION

Recognized: AWE requires a separate mission

The blueprint is amended: AWE™ is **not a processing toggle on daytime imagery**. Three explicit mission products exist:

Product 1 · Stratex Core DayScan™

- **When**: daytime, dry conditions
- **Captures**: RGB imagery only
- **Produces**: measurements, roof/exterior geometry, damage imagery, Digital Twin (Reality + CAD layers), material quantities, DayScan report
- **Updates**: Property Passport DayScan chain

Product 2 · Stratex AWE™ Scan

- **When**: nighttime, only when environmental eligibility passes:
- ΔT between indoor and outdoor ≥ 15°F (configurable per region)
- No rain within last 4 hours
- Wind < 15 mph
- Humidity < 80%
- No direct solar loading (≥1 hour past sunset)
- **Captures**: radiometric thermal + reference RGB
- **Produces**: Air / Water / Energy findings with false-positive controls (moisture-mimic exclusion, ventilation-shadow rejection)
- **Requires**: field verification for any finding above a materiality threshold before Passport entry
- **Updates**: Property Passport AWE chain

Product 3 · Stratex Elite™ Property Intelligence

- Combines DayScan + AWE Scan (with cross-registration between RGB and thermal datasets)
- Produces unified evidence graph, combined report, **AWE Index™** (composite Air/Water/Energy)
- Updates Property Passport with a single approved bundle referencing both scan hashes

Mission planner enforcement

The Mission Planner (§3 blueprint) must present the operator with a **product selector** as the first step: DayScan vs AWE Scan vs Elite. The 20-item ATC gate then loads the product-specific validation subset.

Q6 — DIGITAL TWIN ARCHITECTURE

Canonical decisions

CONCERN	DECISION
Coordinate system	WGS84 lat/lon + local ENU (East-North-Up) origin per property; twin is stored in local ENU with a lat/lon origin pin
Geometry source	Photogrammetry point cloud + mesh (from RGB) — **AWAITING INTEGRATION**
RGB texture source	DayScan RGB assets
Thermal-to-RGB registration	Feature-based homography per capture; per-pixel registration confidence stored
Layer manifest	JSON manifest listing each layer (Reality, CAD, Intelligence) with URIs, hashes, version
Formats	Storage: glTF 2.0 (`.glb`) canonical; IFC 4 and DXF for CAD; PLY for point clouds; radiometric TIFF for thermal source
Measurement provenance	Every extracted measurement carries `derived_from: [asset_ids]` + tolerance ± value
Level-of-detail	glTF LOD hierarchy: L0 (full), L1 (medium), L2 (mobile); browser streams L2 by default, promotes to L0 on zoom
Version comparison	Two twin versions can be diffed; findings that shift receive a `DIFF` event
Browser streaming	glTF over HTTPS with Draco compression + KTX2 textures
Mobile performance	L2 mesh + 1k textures target < 20 MB total; renders on iPhone 12 and later at 60 fps
Vendor-neutral storage	S3-compatible object storage. No vendor-specific format inside the canonical bundle
Exportability	Every twin can be exported as `.glb` + `.ifc` + `.dxf` bundle to a customer download link
External processor replacement	Photogrammetry, CAD/BIM, and radiometric processing are all **adapter contracts**. Concrete provider (Pix4D, RealityCapture, OpenDroneMap) is swappable without changing Core

Stratex-owned vs external-processor-owned

Stratex owns forever:

- Mission package (plan, waypoints, targets)
- Evidence graph (which asset produced which finding)
- Findings + approvals + Passport records
- The layer manifest and its version history
- Signed measurement provenance

External processors own only the raw mesh/CAD generation step and are treated as replaceable adapters behind a `PhotogrammetryProvider` / `BimProvider` interface.

Q7 — DATA-STATE HONESTY

Adopted implementation-state vocabulary

Every feature, integration, and displayed value shall carry exactly one of:

STATE	MEANING
Operational	Live end-to-end in production, verified by tests
Partially operational	Some flows live, others use fallback or fixtures — clearly labeled which
Mocked	Fixture data; nothing real behind it
Planned	Design complete, not built
Awaiting credentials	Built but requires API key / license not yet obtained
Awaiting hardware	Built but requires drone / sensor / dock not yet on hand
Awaiting validation	Built but requires field calibration before trust
Deprecated	Phasing out; do not use for new work
Legacy	Frozen, read-only, under `/legacy/*`

Marketing rules

- No screen may claim "operational" for a component that is any other state
- The UI shall render each state with a distinct chip color and label
- Reports shall footer with a state summary of every value in the document

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Q8 — EXISTING ASSET PRESERVATION MAP

ASSET	DECISION	RATIONALE
Official Stratex branding (logo, colors, typography)	**Preserve unchanged**	Approved; carried into NextGen shell
Property Passport ledger (hash-chained)	**Preserve unchanged**, migrate storage to `passport_ledger` collection with Pydantic model	Working crypto — do not touch the algorithm
Claim Snapshot engine	**Refactor** into `findings` + specialized `claim_snapshot` view; keep public API	Genuine value; needs normalized storage
Adjuster co-sign SHA-256 receipt	**Preserve** as the canonical template for all future human sign-offs	Generalize the pattern to every Human QA event
Open-Meteo weather integration	**Preserve unchanged**; extract into `WeatherProvider` adapter	Simple, working, free
Live Ops WebSocket bus	**Refactor** onto formal event bus (§7 blueprint); keep client contract	WS is fine; server logic needs event log
PDF / Report pipeline (Playwright + cache-first)	**Preserve**; graduate to Report Studio in Delivery Service	Production-safe already
Reports Binder (PyMuPDF)	**Preserve** as embedded viewer in Report Studio	Works
Existing authentication (JWT + bcrypt, CEO login)	**Refactor** — extract to Identity Service, expand to full RBAC	Foundation good; scope must grow
Existing authorization	**Refactor** — introduce `require_auth` + `require_org` on all product routes	Critical gap today
Service credentials / HMAC patterns	**Preserve** patterns; regenerate secrets on each service split	Reusable
Existing Core → Passport → Habitat contracts	**Migrate** — Passport becomes standalone; Habitat contract is new	Formalized in Q2
Existing test suites (`test_iter17_new_features.py` … `test_iter21_cosign.py`)	**Preserve** as legacy regression harness; add NextGen suite alongside	Keep the safety net
Static illustrations in `/twin/*` and `_binder_cache/*`	**Retire** once real twin renders exist, but keep for the interim label as `DEMO DATA`	Cannot be shown as real intelligence
`_sample_analysis()` hardcoded fixture	**Retire** — replaced by the AI Workforce pipeline	Explicit deletion at Phase 3

Nothing is deleted through omission. Every asset is accounted for.

Q9 — SECURITY SCOPE (phased introduction)

Phase 1 (Foundation)

- **Initial roles**: `admin`, `contractor`, `pilot`, `inspector`, `homeowner`, `adjuster`
- **Initial permissions**: read/write/approve/sign
- **Organization isolation**: `org_id` required on every mutable resource; enforced in query builder
- **Property-level access**: `property_org_bindings` table; contractors see only their org's properties
- **Contractor access**: read all properties in their org, write findings after inspector approval
- **Homeowner access**: read-only projection of their property only
- **Pilot / operator access**: read mission plans, write telemetry + capture assets
- **Administrator access**: full within their org; cross-org requires service token
- **Engineer/adjuster reviewer access**: read + sign findings/reports via magic-link tokens (existing cosign pattern)

Service-to-service authorization

- mTLS between Core ↔ Passport ↔ Habitat (Phase 2+)
- Short-lived service tokens with per-endpoint scopes

Audit logging

- Every mutation → `audit_events` (actor, action, resource, before, after)
- Retention: 2 years hot, 7 years cold storage — **AWAITING INTEGRATION**

Secrets management

- All secrets in `.env` today
- Move to Emergent-managed secret store or HashiCorp Vault (Phase 5)

Rate limiting

- Per-user + per-IP token buckets on all public endpoints
- Passport, cosign, storm APIs: 60 req/min per IP; 300 per authenticated user

File access

- All object-storage assets served via **signed URLs** with expiry (default 5 min)
- Signed uploads via presigned POST
- Every download logged to `asset_provenance`

Explicit Phase 1 stops

- No SSO in Phase 1
- No field-level encryption in Phase 1 (Phase 3)
- No SIEM in Phase 1 (Phase 5)

Q10 — DATABASE-DOMAIN JUSTIFICATION

Correction: the 10 domains described in §4 of the blueprint are **logical modules**, not 10 separate databases. During prototype and MVP phases, all 10 domains live inside a **single MongoDB database** with strict per-collection schemas and boundaries enforced at the application layer.

Per-domain table

DOMAIN	PURPOSE	CORE ENTITIES	OWNERSHIP BOUNDARY	TRANSACTIONS	RETENTION	SENSITIVITY	SEPARATE DB?
Identity & Access	Auth + RBAC + audit	orgs, users, roles, api_keys, audit_events	Identity Service	ACID-lite (Mongo multi-doc tx)	7 yrs (audit)	HIGH (PII)	No — logical only
Property Registry	Canonical property record	properties, addresses, parcels, owners, contractors	Property Service	Multi-doc tx on write	Permanent	MEDIUM (PII)	No
Mission Operations	Mission lifecycle	missions, mission_plans, mission_validations, mission_telemetry, capture_sessions	Mission Ops Service	Sequential per mission	3 yrs	LOW-MEDIUM	No
Capture Assets	Binary asset metadata (binary in S3)	assets, asset_derivatives, asset_provenance	Vision Service	Metadata only	7 yrs (bodies in S3)	MEDIUM	No (metadata) / Yes (S3 binaries)
Intelligence Findings	AI + human findings	findings, finding_evidence, finding_confidence, agent_runs	Vision + AWE + Estimating Services	Append-only ledger-like	Permanent	MEDIUM	No
AWE™ Metrics	AWE readings + scores	awe_readings, awe_scores, awe_history	AWE Service	Sequential per property	Permanent	LOW	No
Estimating	Materials + labor + estimates	materials_catalog, labor_catalog, estimates, estimate_versions	Estimating Service	Versioned; last-write-wins	Permanent	LOW-MEDIUM (business)	No
Reports & Deliverables	Rendered reports	report_templates, reports, report_versions, report_signatures	Delivery Service	Sequential	Permanent	LOW	No
Passport & Habitat Sync	Sync audit + queue	passport_ledger, passport_sync_log, habitat_sync_log, sync_retry_queue	Passport Service	Ledger append + idempotent queue	Permanent	HIGH (cryptographic)	**Yes** — Passport becomes standalone in Phase 2
Compliance & Observability	Provenance, events, cost	provenance_chain, event_log, error_events, cost_ledger	Platform	Append-only	2 yrs hot	LOW	No

Only the Passport domain justifies extraction into a separate database (Phase 2) — because its records are the immutable source of truth for all downstream systems. Everything else remains logical modules inside the primary Mongo database until scale demands otherwise.

Q11 — API-PLANE JUSTIFICATION

Correction: the 5 planes are **logical route-prefix conventions with distinct middleware chains**, not 5 separate network services. All planes run inside the same FastAPI app initially; some may split later.

Per-plane table

PLANE	PREFIX	CONSUMERS	AUTH	AUTHZ	VERSIONING	RATE LIMITS	ERRORS	IDEMPOTENCY	AUDIT	NETWORK BOUNDARY?
Public Read	`/api/pub/v1/`	Anonymous (via passport hash)	Signed URL / hash	Resource-level	URL versioned	60/min/IP	RFC 7807 JSON	GET only	Read log to `audit_events`	No (same process, edge-cacheable)
Product	`/api/v1/`	Authenticated users	JWT	RBAC + ABAC (org, property, role)	URL versioned	300/min/user	RFC 7807	Idempotency-Key header on POST/PUT	Full audit	No (same process)
Agent	`/api/agents/v1/`	Backend agent runners	Service token	Scope-per-endpoint	URL versioned	Internal only	RFC 7807	Idempotency-Key required	agent_runs table	**Yes** — internal network only, blocked at ingress
Sync	`/api/sync/v1/`	Passport, Habitat services	mTLS + service token	Service allow-list	URL versioned	Bounded by queue depth	RFC 7807 + retry semantics	Idempotency-Key required	Full audit	**Yes** (Phase 2+)
Webhook	`/api/hooks/v1/`	External inbound (DJI, FAA, weather)	HMAC signature	Provider allow-list	URL versioned	Provider quotas	202 on accept, HMAC failure → 401	Idempotency by event id	Full audit	No (same process)

Actual network boundaries introduced only where necessary — Agent and Sync are gated at the ingress layer. Everything else is logical.

Q12 — EVENT-DRIVEN ARCHITECTURE (Phase 1 events only)

Phase 1 minimum viable event set

Only these events are wired in Phase 1. All others are deferred until real workflows demand them.

EVENT	PRODUCER	CONSUMER	PAYLOAD	IDEMPOTENCY KEY	RETRY	FAILURE	DLQ	AUDIT	SYNC ALTERNATIVE?
`mission.planned`	Mission Ops	Mission Validation	mission_id, plan	mission_id	3× exp backoff	Log + surface in Mission Ops UI	Yes	audit_events + event_log	Sync is fine; keep async
`mission.validated`	Mission Validation	Mission Ops (unlock launch)	mission_id, pass/fail, gate_details	mission_id	none	Block launch	No	audit_events	Sync (immediate)
`capture.received`	Vision (adapter)	Vision Service	mission_id, asset_bundle_id	asset_bundle_id	3×	Escalate to Human QA	Yes	audit_events + event_log	Async needed (large uploads)
`finding.produced`	Any AI agent	Human QA	finding_id, agent_id, confidence	finding_id	none	Halt Passport append	No	agent_runs	Sync (blocks Passport)
`finding.approved`	Human QA	Passport Sync	finding_id, signer, receipt	finding_id	none	Halt sync	No	audit_events	Sync (blocks sync)
`passport.appended`	Passport Service	Habitat Sync	passport_id, seq, receipt_hash	(passport_id, seq)	3×	sync_retry_queue	Yes	passport_sync_log	Async (external system)
`habitat.received`	Habitat (webhook back)	Passport Sync	passport_id, seq, ack	(passport_id, seq)	—	Alert ops	Yes	habitat_sync_log	Async (ack)

No full event-bus infrastructure in Phase 1. These events are recorded in `event_log` and dispatched via in-process async tasks (FastAPI background tasks). If cross-service messaging becomes necessary in Phase 2+, we upgrade to a proper broker (NATS or Redis Streams) — not before.

Q13 — TECHNOLOGY DECISIONS (justified)

TECH	PROBLEM SOLVED	WHY EXISTING IS INSUFFICIENT	ADOPTION COST	MIGRATION RISK	SIMPLER ALTERNATIVE	NECESSARY IN PHASE
React Query	Server-state caching, retries, background refresh, optimistic UI	Current app uses ad-hoc `useEffect` + `fetch`; race conditions and stale data are already visible	Low (drop-in; ~1 day)	Low (per-page adoption)	Continue with useEffect (rejected — bugs)	Phase 1b
Zustand	Small global UI state (e.g. current mission, active twin layer)	React Context works but rerenders too much	Very low	None	Redux (too heavy); Context (already present)	Phase 1b
Event-driven backbone	Decoupling AI pipeline stages	Current code is deeply coupled (`storm_watcher` reaches into passport internals)	Medium	Medium	Sync in-process only (acceptable in Phase 1)	**Deferred to Phase 2**
Blue/green deployment	Zero-downtime cutovers	Emergent single-container deploys create brief unavailability	Low (Emergent may not support natively)	Low	Just rollback via checkpoint (fine for now)	**Deferred to Phase 5**
Feature flags	Ship stubs safely, control AWAITING INTEGRATION states	Currently commenting out code	Low (implement in-house)	None	Env vars (works but less granular)	Phase 1b (in-house key-value flag)
Cryptographic receipts	Prove what was submitted / approved	Already implemented (cosign) — extend the pattern	Free (already exists)	None	(already chosen)	Phase 1 (existing)
RBAC	Role-based access control	Current app has none on product routes	Medium (~1 week)	Medium (retrofit every route)	ACLs only (too rigid)	Phase 1 mandatory
ABAC overlay	Property/org-level access rules	RBAC alone can't gate by `property_org_binding`	Low on top of RBAC	Low	RBAC-only (rejected — leaks data)	Phase 2

Rejected in this round

- **Kubernetes multi-cluster**: unjustified until scale demands
- **Kafka**: overkill; NATS/Redis Streams sufficient when we need a broker
- **GraphQL**: REST + tight schemas is sufficient
- **Rust rewrite of any component**: no
- **Any new frontend framework**: React is the standard here

Q14 — PHASE 1 SCOPE CONTROL (explicit)

Phase 1 **creates the architecture and operating shell** but delivers **no finished production intelligence engines**.

Phase 1 SHALL establish (deliverable list)

1. Application boundaries (Production / NextGen / Legacy)
2. Navigation (14 modules, canonical routes)
3. Workflow state machine (14-stage mandatory chain)
4. Mission shell (Mission Ops workspace stubs)
5. Property shell (Property Portfolio + Property Detail with tabs)
6. Evidence schema (asset ↔ finding ↔ approval)
7. Provenance schema (chip component + backing data model)
8. Agent-run audit schema (`agent_runs` collection)
9. Passport contract (Q2 above)
10. Habitat synchronization contract (§8 blueprint + Q2)

11. Digital Twin layer contract (Q6)
12. AWE mission contract (Q5)
13. Design-system foundation (tokens, provenance chip, module frame)
14. Security foundation (RBAC skeleton, `require_auth` middleware)
15. Test strategy (regression harness plan)

Phase 1 SHALL NOT claim to deliver

- Production-ready photogrammetry
- Production-ready thermal diagnosis
- Automated CAD/BIM generation
- Field-validated AWE conclusions
- Automated damage classification with contract-grade accuracy
- Any measurement presented as production-truth without human sign-off

Every screen in Phase 1 delivering intelligence content shall wear one of these labels:

`DEMO DATA · MOCKED · AWAITING INTEGRATION · AWAITING HARDWARE · AWAITING VALIDATION`.

Success criteria (Phase 1 exit gate)

- All 14 items above delivered
- Blueprint + Appendix + Executive Review signed off
- Screenshots of every workspace captured on desktop + mobile
- Regression suite green
- Zero fabricated intelligence displayed in the shell

APPENDIX — REVISION HISTORY

VERSION	DATE	AUTHOR	CHANGE
1.0	Feb 26, 2026	TC	Initial draft of NextGen Architecture Blueprint (623 lines)
1.0-app	Feb 26, 2026	TC	Review Appendix v1.0 answering Executive Review Questions 1-14

End of Review Appendix.